Setting Up and Running Quantum Espresso with Singularity on GPU

1. Pull the NVIDIA Container

singularity pull docker://nvcr.io/hpc/quantum_espresso:qe-7.3.1

2. Creating pseudos directory and download an UPF file to execute for example

mkdir pseudos

wget "https://archive.materialscloud.org/record/file?record_id=1732&filename=SSSP_1.3.0_PBE_efficiency.tar.gz" -O SSSP_1.3.0_PBE_efficiency.tar.gz

tar -xvzf SSSP_1.3.0_PBE_efficiency.tar.gz -C pseudos/

cd pseudos/

wget https://pseudopotentials.quantum-espresso.org/upf_files/Fe.rel-pbe-spn-rrkjus_psl.1.0.0.UPF

cd ..

mkdir quantum

cd quantum/

vim input.in (the pseudo dir should be replaced with pseudos directory that you created)

```
&control
    calculation = 'scf'
   pseudo_dir = '/home/karim.bezine-ext/espresso/pseudos',
   prefix = 'fe'
   outdir = './tmp/'
/
&system
    ibrav = 2
```

celldm(1) = 10.0

nat = 1

ntyp = 1

noncolin = .true.

lspinorb = .true.

ecutwfc = 30 ecutrho = 300

/

&électrons

conv_thr = 1.0d-8

mixing_beta = 0.7

/

ATOMIC_SPECIES

Fe 55.845 Fe.rel-pbe-spn-rrkjus_psl.1.0.0.UPF

ATOMIC_POSITIONS

(angstrom)

Fe 0.000000 0.000000 0.000000

K_POINTS automatic

4 4 4 0 0 0

Your input.in should be like this :

```
&control
   calculation = 'scf'
   pseudo_dir = '/home/karim.bezine-ext/espresso/pseudos',
   prefix = 'fe'
   outdir = './tmp/'
/
&system
   ibrav = 2
   celldm(1) = 10.0
   nat = 1
   ntyp = 1
   noncolin = .true.
   lspinorb = .true.      ! Enable spin-orbit coupling
   ecutwfc = 30
   ecutrho = 300
/
&electrons
   conv_thr = 1.0d-8
   mixing_beta = 0.7
/
ATOMIC_SPECIES
Fe  55.845 Fe.rel-pbe-spn-rrkjus_psl.1.0.0.UPF
ATOMIC_POSITIONS (angstrom)
Fe  0.000000  0.000000  0.000000
K_POINTS automatic
4 4 4 0 0 0
```

3. Allocating a gpu node then launch the script

salloc -t 7:00:00 -n 1 -A sw_stack-373lcd9r8io-default-gpu -p gpu --gres=gpu:1

vim run_qe.sh :

#!/bin/bashINPUT_FILE="input.in"

NP=1

SIF_PATH="../quantum_espresso_qe-7.3.1.sif"

singularity exec --nv -B "${PWD}:/host_pwd" --pwd /host_pwd "$SIF_PATH" mpirun -np "$NP" pw.x -in "$INPUT_FILE"

chmod +x run_qe.sh

Your script should be like that :

```
#!/bin/bash

INPUT_FILE="input.in"

NP=1

SIF_PATH="../quantum_espresso_qe-7.3.1.sif"

singularity exec --nv -B "${PWD}:/host_pwd" --pwd /host_pwd "$SIF_PATH" mpirun -np "$NP" pw.x -in "$INPUT_FILE"
```

./run_qe.sh

After running you should have an output like this :

```
     cegterg      :     16.06s CPU     16.45s WALL (     637 calls)

     Called by *egterg:
     cdiaghg      :      1.28s CPU      1.43s WALL (    3270 calls)
                                        1.41s GPU  (    3270 calls)
     h_psi        :     11.31s CPU     11.67s WALL (    3278 calls)
                                       11.65s GPU  (    3278 calls)
     s_psi        :      2.17s CPU      2.22s WALL (    3278 calls)
     g_psi        :      0.01s CPU      0.04s WALL (    2633 calls)
                                        0.02s GPU  (    2633 calls)

     Called by h_psi:
     h_psi:calbec :      0.09s CPU      0.19s WALL (    3278 calls)
                                        0.17s GPU  (    3278 calls)
     vloc_psi     :     10.74s CPU     11.06s WALL (    3278 calls)
                                       11.04s GPU  (    3278 calls)
     add_vuspsi   :      0.16s CPU      0.21s WALL (    3278 calls)
                                        0.18s GPU  (    3278 calls)

     General routines
     calbec       :      0.09s CPU      0.12s WALL (    3910 calls)
     fft          :      0.48s CPU      0.56s WALL (    1986 calls)
                                        0.07s GPU  (    1666 calls)
     ffts         :      0.09s CPU      0.10s WALL (     636 calls)
                                        0.01s GPU  (     316 calls)
     fftw         :      1.50s CPU      3.42s WALL (  139772 calls)
                                        2.52s GPU  (  139772 calls)
     interpolate  :      0.57s CPU      0.59s WALL (     320 calls)

     Parallel routines

     PWSCF        :     30.01s CPU     31.06s WALL


   This run was terminated on:  13:57: 3   9Apr2025

=------------------------------------------------------------------------------=
   JOB DONE.
=------------------------------------------------------------------------------=
Warning: ieee_divide_by_zero is signaling
Warning: ieee_inexact is signaling
FORTRAN STOP
[karim.bezine-ext@slurm-a100-gpu-h22a2-u22-sv quantum]$
```